



**UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION**

COURSE	:	DATABASE DESIGN AND DEVELOPMENT
COURSE CODE	:	ICT450 / ITS432
EXAMINATION	:	FEBRUARY 2025
TIME	:	3 HOURS

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of FIVE (5) questions.
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of:
 - i) the Question Paper
 - ii) an Answer Booklet – provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 5 printed pages

QUESTION 1

- a) Data is a raw fact or a building blocks of information. Identify **FOUR (4)** examples of data for an online store.
(4 marks)
- b) Describe **TWO (2)** differences between a database and a spreadsheet.
(4 marks)
- c) There are three rules of referential integrity which are Insert Rule, Update Rule and Delete Rule. Describe these three rules and illustrate your answer.
(6 marks)
- d) Database system is designed based on two design strategies. Explain **THREE (3)** differences between these two designs.
(6 marks)

QUESTION 2

- a) The evolution of data models has been driven by the increasing complexity of data storage, management, and analysis needs over time. Describe **TWO (2)** disadvantages of hierarchical model.
(4 marks)
- b) Business rules are precise statements, derived from a detailed description of the organization's operations. Identify **FOUR (4)** sources of business rules.
(4 marks)
- c) Backup the database is one of the crucial aspects in Database Life Cycle (DBLC). Describe **THREE (3)** types of backup.
(6 marks)
- d) Briefly explain **THREE (3)** advantages of mobile database in marketing area.
(6 marks)

QUESTION 3

Table 1 shows a daily order for a new start-up company, Ridz's Burger. This order is manually recorded in a logbook. Each customer can make several orders and each order is managed by one staff.

Table 1: Ridz's Burger Logbook's Record Order

Date	Name of Customer	Total Orders	List of orders and Quantity	Total payment	Type of Payment	Name of Staff
1/2/2025	Intan	2	B1 - 1 CK2 - 1	RM 6.50	QR Code	Ridz
1/2/2025	Azwani	1	CK1 - 1	RM 4.00	Cash	Ridz
2/2/2025	Lya	1	BF3 - 1	RM 5.00	Cash	Syafinaz
4/2/2025	Lutfi	3	B1 - 3	RM 6.00	QR Code	Syafinaz

Indication

B1 – Benjo

CK1 – Chicken Burger

CK2 – Chicken Cheese Burger

BF1 – Beef Burger

BF2 – Beef Cheese Burger

BF3 – Beef Special Burger

Based on the information given, answer the following questions:

- State **THREE (3)** problems that can be happened to the data in this logbook's record.
(3 marks)
- Based on the scenario in Table 1, create the primary key(s) and identify partial dependencies and/or transitive dependencies.
(6 marks)
- Normalize this logbook's record from the First Normal Form (1NF) until the Third Normal Form (3NF). List all the steps involved.
(11 marks)

QUESTION 4

The relational schema below shows the e-Travel Package database system for one of the local travel agents in Malaysia.

CUSTOMERS (<u>CustomerID</u>, FirstName, LastName, PhoneNumber) BOOKING (<u>BookingID</u>, BookingDate, NumberOfPax, TripDate, CustomerID*, GuideID*, PackageID*) GUIDE (<u>GuideID</u>, GuideName, BasicSalary, HireDate) PACKAGE (<u>PackageID</u>, PackageName, PricePerPax, PackageDetails)

Write the SQL statements based on the following requirements:

- a) Calculate the total number of trips for each date.
(4 marks)
- b) Display the details of customer who make a highest number of booking.
(4 marks)
- c) Increase 10% of the salary for all guides who works on a trip in February 2025.
(4 marks)
- d) Display the full name of packages that has been booked more than 70 times.
(4 marks)
- e) Identify the total number of customers that make a booking in Feb 2025 until March 2025.
(4 marks)

QUESTION 5

Excellent Art is a company that provide art training for children who are interested in it. The company mission is to promote creative and innovative thinking in children's life.

- There are about 15 Excellent Art studios around Malaysia. To uniquely identify each art studio, they had been assigned with a unique studio id. Other important information that needs to be kept in the database are studio name, address, phone number, email, and GPS coordinate.
 - Each studio deals with many suppliers, which supply a variety of art materials. Art material information need to be recorded are art material id, name, description, and price. The information about the supplier that must be recorded is its unique id, name, address, and email.
 - One art studio provides many programs. Each program will uniquely have identified by program id, name, description, and price. The programs are classified to Visual Art, Glass Painting and Handicraft. Each of the program will have their own additional information, such as, Visual Art program will have class type, Glass Painting program will have program level, with value 1 to 5 as it will indicate the technique in glass painting and Handicraft program will need craft data.
 - There are about 100 employees all together in the Excellent Art company. Each employee has a unique employee number, first name, last name, job titles and date of birth and is allocated to only one art studio. Each studio has several employees. However, each employee only can work in one studio.
 - If an employee currently married to another employee of the same studio, the spouse's employee number and date of marriage is also recorded.
 - To be creative, many employees in the studio acquire many skills. Skills data to be recorded are skill type and skill description.
- a) Construct an Entity Relationship Diagram (ERD) complete with all the required tables, attributes, relationships as well as the cardinalities using the Crow's Foot notation that reflects all the given business rules.

(16 marks)

- b) List **TWO (2)** example of reports that can be produced from this database system.

(4 marks)

END OF QUESTION PAPER